

Architecture — how the pieces connect

Two codebases — read this before anything else

	russell-capital-systems/	russell-capital/
TypeScript files	502	80
Shared engines	125	20
Test files	201	11
package.json / router	yes / yes	no / no
Imported by the deployed app	—	nothing imports it

russell-capital-systems/ is the deployed app. russell-capital/ is a staging folder that looks like an app and is not one: ~5,900 lines of finished, tested engine code that executes nowhere, including an adverse-scenario stress tester, a full property-and-loan amortisation engine, an illustration-compliance layer and a small-business lending vertical. Four module names exist in **both** folders and have diverged.

Before writing any new engine, search both folders for its name. The file-by-file correlation and the integration order are in master_build_correlation.md; the machine-readable version is data/master_build_inventory.csv.

The stack

Express + tRPC 11 server · Vite/React 19 client with wouter · Drizzle ORM on MySQL · Vitest · Tailwind 4. Railway deploys from master. Shared TypeScript modules under shared/ hold every engine so the server, the client and the tests run the same arithmetic.

The façade layer (added this session)

Six domains, each reachable in one call. Two exist:

- realEstateCapitalStackEngine.ts — capitalStack(situation) returns open and blocked mechanisms with reasons, the thresholds that actually bind, the legal-plan count at a fixed depth, ranked plans, and the nearest archetype with a closeness score.
- historicalMarketRegimeEngine.ts — regimeAt(year), conditionalOdds(series, horizon, bucket) carrying the window count, powerSwing() returning null rather than a number when a bucket is thin, and coverage().

Four are still owed: taxEngine, insuranceEngine, householdEngine, careerEngine. The façade pattern is what makes the system addressable by an AI channel, a partner API or a mobile client without knowing five module names and their call order.

The one object

Situation (shared/sequencePlanner.ts) is the household: rentals, values, LTVs, rent, primary residence, surplus, reserve, documentation, goal, horizon, insurability, off-market access, uplift, appreciation, rate. Every engine should take it or a projection of it. Today the fact finder, the genome and the planner each hold their own shape; making

them one is P1 task 1.9 in recommendations.md.

Data flow

```
Fact finder (shared/clientFactFinder.ts)
  ■■ Situation
    ■■ Wealth Genome (wealthGenomeFactors, wealthGenomeDurability)
    ■ ■■ Household Genome (householdGenome: combination rules, influence, veto, consent)
    ■■ Strategy Fit (genomeStrategyFit over genomeStrategies: signals, gates, silence ≠ evidence)
    ■■ Sequence Planner (sequencePlanner: moves, per-asset legality, thresholds → stages)
    ■ ■■ cycleEngine (mechanism parameters, simulateCycle)
    ■ ■■ sequenceOrderings (roles, hard rules, plausibility)
    ■ ■■ thresholds (standards, variants, fixed)
    ■ ■■ sequenceArchetypes (named presets)
    ■■ Provenance (provenance: FigureTrace per headline figure – 3 exist, 31 featured pages need one)
    ■■ Memory bank (aiMemoryBank: 29 groups → compositeMind → twelve channels)
      ■■ channels answer only with a source and name pages by real path
```

Registries and the tests that guard them

Registry	Guarded by	Fails when
calculatorCatalog.ts (115 pages)	server/calculatorCatalog.test.ts	a path has no route (literal or :param), a page file is missing, an engine is named that is not on disk, a route is declared twice
aiMemoryBank.ts (29 groups)	server/cycleScenarios.test.ts	a module named in a group is not on disk; a group is unwired
altCredit/lenders.ts (15 records)	server/altCredit.test.ts	any phone-shaped string outside a verified phone field; a Verified value without source and date
mechanismDossiers.ts (39 providers)	server/mechanismDossiers.test.ts	a registryId with no record; a phone, email, percentage or dollar figure attached to a named company; value and frequency collapse
sequenceOrderings.ts (42 orderings)	server/sequenceOrderings.test.ts	an annotated ordering is illegal; shares + residual ≠ 100; the hand-picked top ordering lands in the bottom half by the independent plausibility score
sequenceArchetypes.ts (12)	server/sequencePlanner.test.ts	any archetype's sequence is refused by the planner against its own situation
thresholds.ts (14)	same	a figure without a source URL; a threshold neither fixed nor movable
pageRatings.ts (52)	server/integrationAudit.test.ts	a rated path not in the catalogue; a featured page unrated
patentStatus.ts	server/patentClaimGuard.test.ts	any surface says "patent pending" while APPLICATIONS is empty
Façades (2)	server/facadeEngines.test.ts	a capital-stack refusal lacks a reason; a regime figure lacks its sample size; powerSwing states a number from a thin bucket
App routes (321)	grok-merge.smoke.test.ts, managed-port.smoke.test.ts	the route count changes without the test being updated

Live data

Twelve tRPC routers fetch external sources on sweeps and store with as-of dates: zip (FHFA, Zillow, PMMS), erosion (FRED CPI, tax history, power feeds), outsideForces (40+ series), career (BLS OEWS, NCES), rental

(FRED, FEMA NRI, Fannie), ltc, incomeLife (SSA), inheritance, forgiveness, taxSchedule, unasked, integration (reads the repo). Pages that show a rate, limit or price and do not call one of these are flagged `liveData: false` on the scorecard — 104 of them today.

The integration audit

`server/integrationAudit.ts` reads the router, the catalogue, each page component's `@shared/` imports, the memory-bank module lists, the test files' imports, the genome's `relatedPaths`, `sphere.ts`, `provenance.ts`, and which routers' files (plus their local imports) reference an external data host. Ten dimensions, weights visible (brain 1.5, sphere 0.5), summing to ten. `taxBracketEngine` imported as a formatter on ~60 pages is deliberately not counted as an engine wire. Served live by `integration.scorecard`; generated to `reports/scorecard.md` by `server/integrationScorecard.script.ts`.

Conventions that hold everything together

1. **Verified<T>** — `{verified:true,value,source,asOf}` | `{verified:false,whyNot,checkAt}`. No third shape.
2. **Silence ≠ evidence** — an unanswered genome factor contributes to neither numerator nor denominator; it lowers confidence.
3. **Gates vs signals** — gates can BLOCK at any fit; `external: true` gates cannot be settled by any genome reading.
4. **Legality per asset** — covenants bind the property they are on; refusals name the clause.
5. **Two numbers, never one** — value/frequency, confidence/likelihood, share/likelihood, wiring/worth. Tests assert they differ.
6. **Amortisation is the dividing line** — obligations that pay themselves down stack; those that don't are counted as balloons and penalised.
7. **Priority trimming is a decision** — memory groups at priority 1 are never trimmed; 3 drops first.

The convention this codebase should adopt

`UnsourcedFindingError` (in `russell-capital/shared/council/aiCouncil.ts`) and `UnsourcedQuoteError` (in `helocLenders.ts`) make an unsourced figure a runtime error that throws. `Verified<T>` plus guard tests make the *shape* hard to get wrong and catch source-text violations at build time — but no test can scan a value that did not exist when the test ran. The two together are belt and braces: the type makes it hard, the test makes it visible, the error makes it impossible. Promote both classes into `shared/sourcing.ts` and route every client-visible number through `assertSourced`.

What to build next, structurally

The universal situation (P1 1.9), then provenance on every featured page (1.2), then genome routing to every rated page (1.1). Those three raise the scorecard mean more than anything else because they are the dimensions with the largest gaps and the ones the other criteria depend on.